Theory Assignment Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	5
Level of Content	1					
Development	2					
Spelling & Grammar	1					
Organization and Formatting	1					
					Total obtained mark	
Comments						

Semester: Summer 25

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Batch: 241 (42) **Section: E**

Course Code: STA 101 **Course Name: Statistics I**

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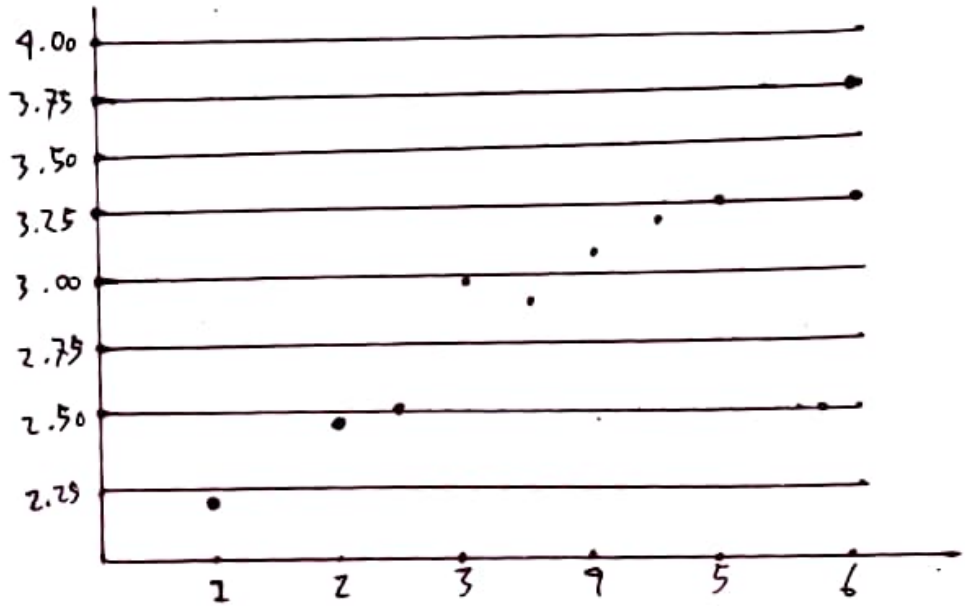
Designation:

Submission Date: 20/08/2025

ans. to the q. No. 1

i



The scatterplot shas strong positive correlation.

ii

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	2.2	-2.7	-0.83	2.241	7.29
2	2.4	-1.7	-0.63	1.071	2.89
2.5	2.5	-1.2	-0.53	0.636	1.44
3	2.8	-0.7	-0.23	0.161	0.49
3.5	3.0	-0.2	-0.03	0.006	0.04
4	3.2	0.3	0.17	0.051	0.09
4.5	3.3	0.8	0.27	0.216	0.64
5	3.5	1.3	0.47	0.611	1.69
5.5	3.6	1.8	0.57	1.026	3.24
6	3.8	2.3	0.77	1.771	5.29

Now,

$$\bar{x} = \frac{37}{10} = 3.7 \; ; \; \bar{y} = \frac{30.3}{10} = 3.03$$

$$\therefore \; (x_i - \bar{x})^2 = 23.1$$

$$(x_i - \bar{x})(y_i - \bar{y}) = 7.79$$

$$\therefore \beta_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{7.79}{23.1} = 0.337$$

$$\beta_1 = \bar{y} - \beta_2 \bar{x} = 3.03 - (0.337 \times 3.7)$$

$$= 1.783$$

Therefore, the model in question is, $\hat{y} = 1.783 + 0.337x$

iii

Here, intercept ($\beta_0 \approx 1.783$) represents the GPA of the student when the study hour is zero. Slope ($\beta_1 \approx 0.337$) represents the change in the GPA for the hour of study per day

iv

x_i	y_i	$(x_i-\bar{x})^2$	$(y_i-\bar{y})^2$	$\{(x_i-\bar{x})(y_i-\bar{y})\}^2$
1	2.2	7.29	0.6889	5.022081
2	2.4	2.89	0.3969	1.147091
2.5	2.5	1.44	0.2809	~~0.259921~~ 0.904996
3	2.8	0.49	0.0529	~~0.000036~~ 0.259921
3.5	3.0	0.09	0.0009	~~0.002601~~ 0.000036
4	3.2	0.09	0.0289	~~0.096156~~ 0.002601
4.5	3.3	0.64	0.0729	~~0.373321~~ 0.096156
5	3.5	1.69	0.2209	~~1.05~~ 0.373321
5.5	3.6	3.24	0.3249	1.052676
6	3.8	5.29	0.5929	3.136441
		$\Sigma = 23.1$	$\Sigma\ 2.661$	$\Sigma = 11.445179$

$$\therefore r^2 = \frac{\Sigma\{(x_i-\bar{x})(y_i-\bar{y})^2\}^2}{\Sigma (x_i-\bar{x})^2 \times \Sigma (y_i-\bar{y})^2}$$

$$= \frac{11.445179}{23.1 \times 2.661}$$

$$= 0.1862$$

$r^2 = 0.1862$ denotes the fact that 18.62% of the variation in GPA is explained by the study hour.

V

Using $\hat{Y} = 1.783 + 0.337x$

since $x = 7$ hours

$$\hat{Y} = 1.783 + 0.337 \times 7$$

$$= 4.14$$

$\therefore$ The GPA for 7 hours per day is 4.14

Ans. to the q. No. 2

Given that,

$$P(D) = \frac{1}{1000} = 0.001$$

True positive rate: $P(\text{Positive} \mid D) = 0.95$

True Negative rate: $P(\text{Negative} \mid -D) = 0.95$

$$\Rightarrow P(\text{Positive} \mid -D) = 0.05$$

$$\therefore P(-D) = 1 - 0.001$$

$$= 0.999$$

By applying Bayes' Theorem,

$$P(D \mid \text{Positive}) = \frac{P(\text{Positive} \mid D) \cdot P(D)}{P(\text{Positive} \mid D) \cdot P(D) + P(\text{Positive} \mid -D) \cdot P(-D)}$$

$$= \frac{(0.95) \times (0.001)}{(0.95) \times (0.001) + (0.05)(0.999)}$$

$$= \frac{0.00095}{0.0509}$$

$$\doteq 0.0187 = 1.87\%$$

If a person tests positive, the probability they actually have the disease is only about 1.87%.

Ans. to the q. No. 3

(i)

x	f	$P(x_i)$
1000	10	$10/100 = 0.1$
1500	20	$20/100 = 0.2$
2000	40	$40/100 = 0.4$
2500	20	$20/100 = 0.2$
3000	10	$10/100 = 0.1$

We convert frequencies to probabilities

$$\therefore P(x) = \frac{\text{Frequency}}{\text{total}}$$

$\therefore$ This is the probability distribution

ii

The expected value (mean) is -

$$E(x) = \sum x_i P(x_i)$$

$$= 1000 \times (0.1) + 1500 \times (0.2) + (2000 \times (0.4))$$

$$+ 2500 \times (0.2) + 3000 \times (0.1)$$

$$= 100 + 300 + 800 + 500 + 300$$

$$= 2000$$

$\therefore$ expectated lifetime 2000 hours

(iii)

x_i	$P(x_i)$	$(x_i)^2$	$(x_i)^2 P(x_i)$
1000	0.1	1000000	100000
1500	0.2	2250000	450000
2000	0.4	4000000	1600000
2500	0.2	6250000	1250000
3000	0.1	9000000	900000

$$\therefore var(x) = E(x^2) - [E(x)]^2$$

$$= \sum x_i^2 P(x_i) - [\sum x_i P(x_i)]^2$$

$$= 4300000 - (2000)^2$$

$$= 300000$$

$$\therefore S.D(x) = \sqrt{var(x)}$$

$$= \sqrt{300000}$$

$$= 547.72$$

(iv)

Probability that a bulb lasts a least 2000 hours

$\therefore P(x \geq 200) = P[x = 2000] + P[x = 2500] + P[x = 3000]$

$= 0.4 + 0.2 + 0.1$

$= 0.7$

v

Let,

$p = P(x \geq 2000) = 0.7$

$q = 1 - p$

$= 0.3$

Use binomial Formula,

$P(x) = {}^5C_3 \cdot (0.7)^3 \cdot (0.3)^2$

$= 10 \times (0.343) \times (0.09)$

$= 0.3087$

$\therefore$ Probability that exactly 3 of 5 last at least 2000 hours 30.87%.

Ans. to the Q. No. 4

i

Let x be the number of patieants cured.

Distribution:

$$x \sim \text{Binomial}(n=8, p=0.7)$$

ii

P [exactly 6 patients are cured]

$$= P(X=x) = {}^{n}C_{x} \cdot p^{x} \cdot q^{n-x}$$

$$\therefore P(x=6) = {}^{8}C_{6} \cdot (.7)^{6} \cdot (.3)^{8-6}$$

$$= 28 \times (.7)^{6} \times 0.09$$

$$= 0.2965$$

iii

P [at least 5 patients are cured]

$$= P(x \geq 5)$$

$$= P(x=5) + P(x=6) + P(x=7) + P(x=8)$$

$$= {}^{8}C_{5} \times (.7)^{5} \times (.3)^{3} + {}^{8}C_{6} \times (.7)^{6} \times (.3)^{2} + {}^{8}C_{7} \times (.7)^{7} \times (.3)^{1}$$

$$\times {}^{8}C_{8} \times (.7)^{8} \times (.3)^{0}$$

$$= 0.2541 + 0.2965 + 0.1977 + 0.0576$$

$$= 0.8059$$

iv

Expected cured patients (mean) $= np$

$= 8 \times 0.7$

$= 5.6$

variance $= npq = 8 \times .7 \times .3$

$= 1.68$

v

The binomial model is appropriate if,

1) Each patient outcome is Bernouli trial (cured/not cured)

2) Trials are independant

3) Probability of success $p = 0.7$ is the same for every patient in the sample.

if any of those ~~assu~~ assumptions fail, the binomial model may not be appropriate.

Ans. to the q. No. 5

a

Here, $H_0 : \mu = 15$

$H_a : \mu > 15$

also, $\sigma = 4.5$ $n = 99$, $\bar{x} = 16.5$, $\sigma = 5\% = 0.05$

since $n = 99 \geq 30$,

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$= \frac{16.5 - 15}{\frac{4.5}{\sqrt{99}}} = 2.33$$

Now, for right tailed ($H_a : \mu > 15$),

$$P_{tab} = P(z \geq 2.33)$$

$$= 0.0099$$

since $z_{cal} > P_{tab}$, we reject H_0 at 5% level of significance

b

$H_0 : \mu = 80$

$H_a : \mu > 80$

Here, $\sigma = 5$, $n = 36$, $\bar{x} = 82$, $\alpha = 0.05$

Since, $n = 36 > 30$, the test statistic,

$$z = z_{cal} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{82 - 80}{\frac{5}{\sqrt{36}}} = 2.4$$

Now, for right-tailed test,

$$z_{tab} = P(z \geq 2.4)$$
$$= 0.0082$$

Since, $z_{cal} > z_{tab}$, we reject H_0 at 5% significance level.